

SOCIAL MEDIA ANALYZER

Introduction

The **Social Media Content Analyzer** is a full-stack web application designed to analyze social media content using **Artificial Intelligence (AI) and Natural Language Processing (NLP)** techniques. It allows users to submit social media text and receive meaningful insights such as sentiment, content classification, keywords, and actionable recommendations. The application provides a simple and responsive interface for presenting the analysis results.

Approach

The project follows a **client-server architecture**. The frontend is developed using **React and Vite**, where users enter content and view the analysis results. The submitted content is sent to a **Python FastAPI backend** through REST APIs. The backend validates and preprocesses the input, performs AI/NLP-based analysis, and returns the results as structured JSON data. The frontend then displays these results through an interactive dashboard.

Environment variables are used for configuration, while CORS enables frontend-backend communication. For deployment, the frontend is hosted as a **Render Static Site**, and the backend is deployed as a **Render Web Service**. This approach ensures a modular, maintainable, and scalable application structure.